

Prashant Arya - AI Engineer

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PROFILE SUMMARY

AI Engineer with 2+ years of experience across Generative AI, LLMs, NLP, and ML/DL, designing and deploying production LLM applications, RAG pipelines, and multi-agent systems. Led the design and deployment of MYA, an enterprise multi-agent chatbot used by 80,000+ Bosch employees, plus HR automation and analytics solutions using Python, FastAPI, Azure, LangChain/LangGraph, AutoGen, and LLM fine-tuning (QLoRA, PEFT).

SKILLS SUMMARY

- **Generative & Agentic AI:** LLM applications (Llama 3.1, Azure OpenAI), Multi-Agent systems (LangGraph, LangChain, AutoGen, CrewAI), RAG, LLM fine-tuning (QLoRA, PEFT), prompt engineering, Hugging Face.
- **ML / DL & NLP:** Classical ML/DL, NLP, TensorFlow, PyTorch, Scikit-learn, spaCy, NLTK, BERTopic, Transformers.
- **Cloud & MLOps:** Azure Cloud (Azure AI Foundry), Docker, Git, MLflow, vector databases (ChromaDB, FAISS).
- **Languages & Tools:** Python, Java, JavaScript, FastAPI, REST APIs, SQL (MySQL, PostgreSQL), Vue.js.

EXPERIENCE

Bosch Global Software Technologies

Hyderabad, India

AI Engineer (Full-time)

Jan 2024 – Present

- Designed and led end-to-end architecture and deployment of MYA, an enterprise GenAI multi-agent LLM chatbot serving 80,000+ employees, using FastAPI, Azure OpenAI, asynchronous microservices, and secure SSO (Azure Entra ID).
- Architected and owned a scalable multi-agent orchestration framework with intelligent intent routing, guarded tool/action execution, automated workflows, and enterprise-grade reliability and safety controls.
- Built a FastAPI backend orchestrating a LangGraph-style RAG pipeline (query understanding → retrieval → generation) using Azure OpenAI LLM models, multilingual query preprocessing (language detection and procurement abbreviation expansion), and structured JSON outputs (answers, citations, quality flags).
- Led the development and deployment of an internal AI-powered web app (FastAPI, Azure, GenAI, LangChain) for HR workflow automation, cutting manual effort by 80% and turnaround time by 75% across teams.
- Led multilingual sentiment and topic analysis using BERTopic, Azure OpenAI, spaCy, and NLTK, delivering actionable workplace insights that reduced HR survey analysis time by 90% and improved employee satisfaction.
- Developed and maintained a time-series forecasting pipeline (Pandas, Nixtla), boosting order line forecast accuracy by 15% while automating data workflows for faster business decisions.

Microsoft

Remote

Future Ready Talent – AI Intern

Dec 2021 – Mar 2022

- Built an AI-powered COVID-19 conversational chatbot using Microsoft Bot Framework, WHO API, and Azure, delivering real-time information via Telegram.
- Designed multi-turn conversational flows and NLP-based intent handling with real-time data integration, collaborating with Microsoft ML/AI experts to meet production-readiness standards.

PROJECTS

- **Multi-Agent AI Sales Assistant (Freelance Project) | GitHub Repository**
 - Developed a **multi-agent sales intelligence platform (Python, LangChain, LangGraph)** to **automate B2B prospecting** with personalized insights and dynamic communication.
 - Implemented a **hybrid AI architecture** combining **Groq LLM** with deterministic Python logic for data filtering, scoring, and SWOT analysis.
 - Built a **CRM pipeline (Pandas)** with SEO/Social scoring, hybrid search tools, and a **Streamlit** chat interface with real-time monitoring and visual analytics.
- **Multi-Agent AI Travel Planner | LangGraph, Groq LLaMA 3.3, FastAPI, Streamlit | GitHub Repository**
 - **Personalized Itinerary Generation:** Built a multi-agent AI travel planner using LangGraph agents and Groq LLaMA 3.3 to generate multi-day itineraries based on user preferences, budget, and group type.
 - **Modular Architecture and UX:** Designed a Python FastAPI backend with strong state management and input validation, and a Streamlit frontend for interactive itinerary creation and accommodation recommendations.

ACHIEVEMENTS

- **Microsoft Certified: Azure AI Engineer Associate** – designing and deploying Azure AI solutions and agents using Azure Cognitive Services, Azure OpenAI, and Azure ML for NLP, computer vision, and generative AI. [View Credential](#)
- **Microsoft Certified: Azure Data Scientist Associate (DP-100)** – Azure Machine Learning for data prep/exploration, model training, ML pipelines/jobs, deployment & monitoring, and MLflow experiment tracking and AI apps using Azure AI services. [View Credential](#)
- **Published Paper: Design of Intelligent Box for Pharmaceutical Assistance for Senior Citizens, ICONIC 2023.**
- **Winner, Innovation Mission Punjab MIC (2023):** healthcare solution assisting senior citizens with medication reminders and health monitoring.

EDUCATION

Bachelor of Technology – Electronics and Communication Engineering | GPA: 7.1
Lovely Professional University

Punjab, India
July 2019 - July 2023